

# Multiple Linear and Nonlinear Regression

## AGRON 5130 - Unit 11 Assignment

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### Introduction

The goal of this assignment is to help you practice the full workflow from Module 11:

- interpreting partial effects in multiple regression
- diagnosing multicollinearity and unstable coefficients
- comparing full vs reduced models
- fitting and interpreting a nonlinear growth model
- connecting model choice to biological reasoning

You will complete two parts:

- **Part 1:** multiple linear regression using soybean yield data
- **Part 2:** nonlinear modeling using tomato growth data

Submit a knitted PDF/HTML generated from your R Markdown file to Canvas.

**Total: 15 points**

### Part 1: Multiple Linear Regression (Thompson Soy)

```
library(ggplot2)
library(car)
```

```
## Loading required package: carData
```

```
thompson_soy <- read.csv("data/thompson_soy.csv")
head(thompson_soy)
```

```
##   yield rain6 rain7 temp6 temp7
## 1    17  3.36  1.01  71.9  79.1
## 2    18  3.19  2.97  75.5  79.3
## 3    20  4.08  3.32  74.0  77.6
## 4    15  1.48  2.41  78.2  78.1
## 5    19  3.01  3.24  79.0  81.7
## 6    18  5.99  3.33  68.2  79.0
```

For this dataset:

- `yield` = soybean yield
- `rain6` = total June precipitation
- `rain7` = total July precipitation
- `temp6` = mean June temperature
- `temp7` = mean July temperature

The goal is to ask which of these weather variables are associated with yield **after accounting for the others**.

## Question 1

1 point

Fit a multiple linear regression model of `yield` as a function of all four weather variables:

- `rain6`
- `rain7`
- `temp6`
- `temp7`

```
model1 <- lm(yield ~ rain6 + rain7 + temp6 + temp7,  
             data = thompson_soy)
```

## Question 2

1 point

Report model coefficients.

```
coef(model1)
```

```
## (Intercept)      rain6      rain7      temp6      temp7  
## 70.64406192 -0.37426431  1.45254138 -0.07801906 -0.64025805
```

Write 2-3 sentences interpreting:

1. one precipitation coefficient as a partial effect
2. one temperature coefficient as a partial effect

Use language like: “Holding the other predictors constant...”

## Question 3

1 point

Report model fit statistics from `summary()`:

```
summary(model1)
```

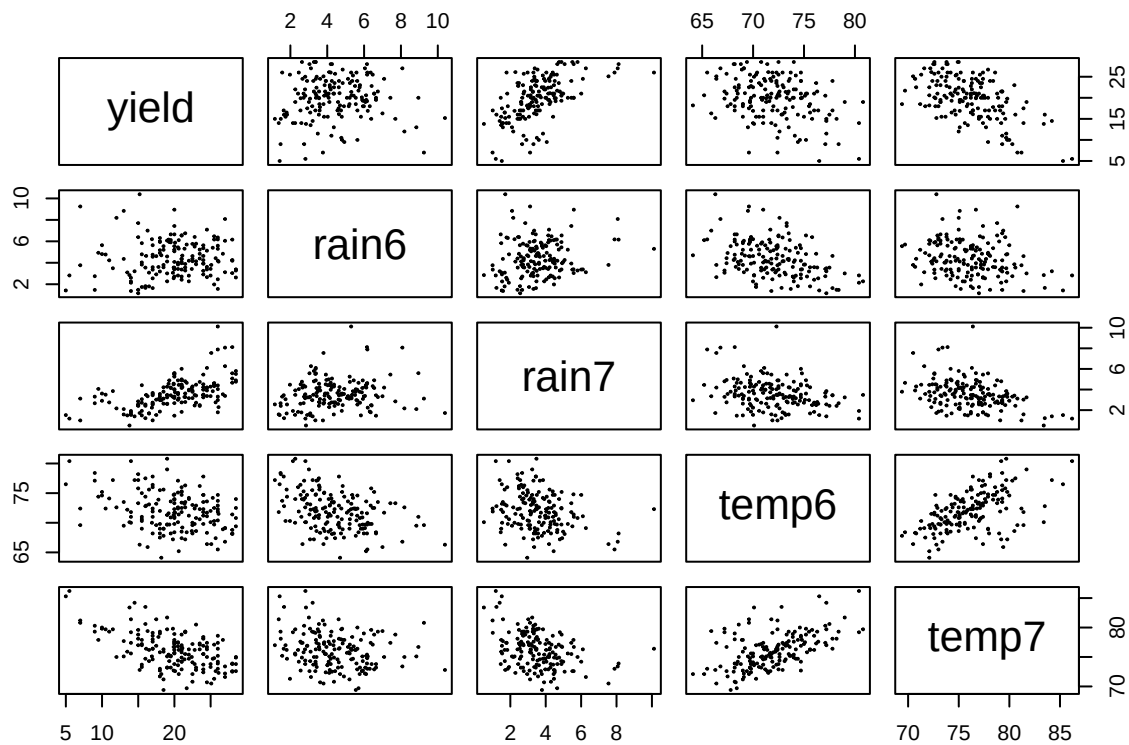
```
##
## Call:
## lm(formula = yield ~ rain6 + rain7 + temp6 + temp7, data = thompson_soy)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.5155 -2.4670 -0.0122  2.7631 10.3798
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  70.64406    9.57142   7.381 8.10e-12 ***
## rain6        -0.37426    0.18672  -2.004  0.0467 *
## rain7         1.45254    0.21915   6.628 4.95e-10 ***
## temp6        -0.07802    0.11309  -0.690  0.4913
## temp7        -0.64026    0.11793  -5.429 2.06e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.701 on 160 degrees of freedom
## Multiple R-squared:  0.4617, Adjusted R-squared:  0.4483
## F-statistic: 34.31 on 4 and 160 DF,  p-value: < 2.2e-16
```

## Question 4

1 point

Create a scatterplot matrix of model variables.

```
plot(thompson_soy, pch = 20, cex = 0.3)
```



Then name one pair of predictors that appears strongly related.

## Question 5

1 point

Create a correlation matrix for the predictor variables only.

```
cor(thompson_soy) |> round(2)
```

```
##      yield rain6 rain7 temp6 temp7
## yield  1.00  0.05  0.56 -0.32 -0.55
## rain6  0.05  1.00  0.17 -0.41 -0.23
## rain7  0.56  0.17  1.00 -0.28 -0.39
## temp6 -0.32 -0.41 -0.28  1.00  0.53
## temp7 -0.55 -0.23 -0.39  0.53  1.00
```

## Question 6

1 point

Describe and interpret the correlations between variables. Are there any variables you would be concerned about including in the model? Why?

## Question 7

1 point

Compute VIF values for the full model.

```
vif(model1)
```

```
##      rain6      rain7      temp6      temp7  
## 1.205412 1.192896 1.601007 1.531871
```

Then answer:

1. Which predictor has the largest VIF?
2. Is multicollinearity likely to affect interpretation?

## Question 8

1 point

Create a reduced model by dropping one predictor you believe is least useful or most redundant.

```
modelr <- lm(yield ~ rain6 + rain7 + temp6,  
             data = thompson_soy)
```

Compare full vs reduced model using:

1. ANOVA (anova(reduced, full))
2. AIC (AIC(reduced, full))

```
anova(modelr, model1)
```

```
## Analysis of Variance Table  
##  
## Model 1: yield ~ rain6 + rain7 + temp6  
## Model 2: yield ~ rain6 + rain7 + temp6 + temp7  
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)  
## 1     161 2595.2  
## 2     160 2191.5  1    403.74 29.477 2.065e-07 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
AIC(modelr, model1)
```

```
##           df          AIC  
## modelr    5 932.9048  
## model1    6 907.0044
```

In 3-4 sentences, justify which model you would keep for interpretation.

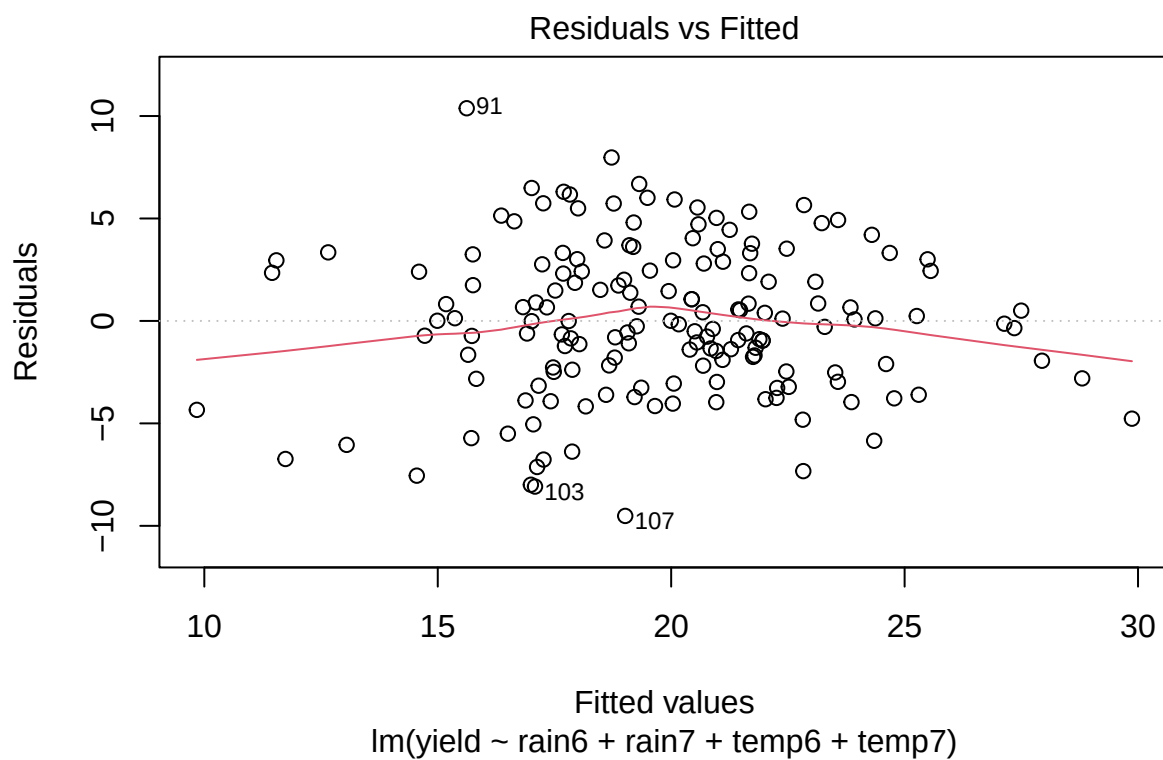
## Question 9

1 point

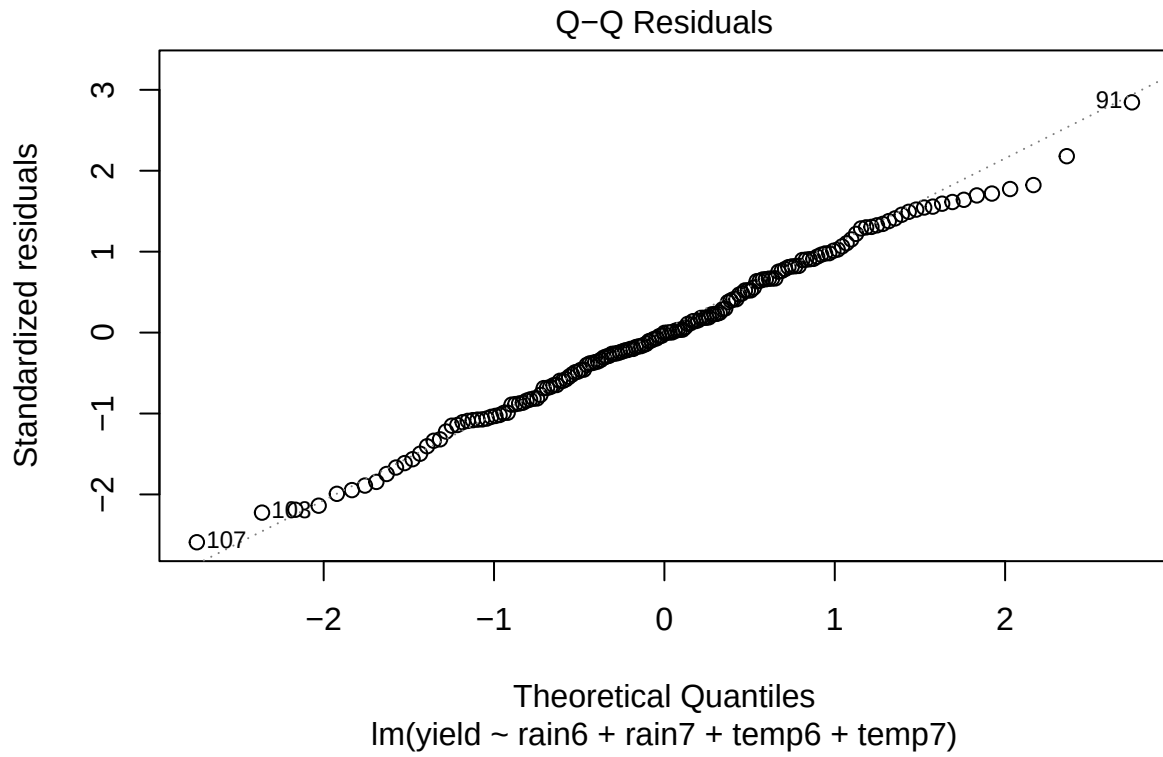
Check assumptions for your **selected** model using:

1. residuals vs fitted plot (`which = 1`)
2. normal Q-Q plot (`which = 2`)

```
plot(model1, which = 1)
```



```
plot(model1, which = 2)
```



Write 2-3 sentences on whether the linear model is adequate.

## Part 2: Nonlinear Regression (Tomato Growth)

```
tomato <- read.csv("data/tomato_days_volume.csv")
head(tomato)
```

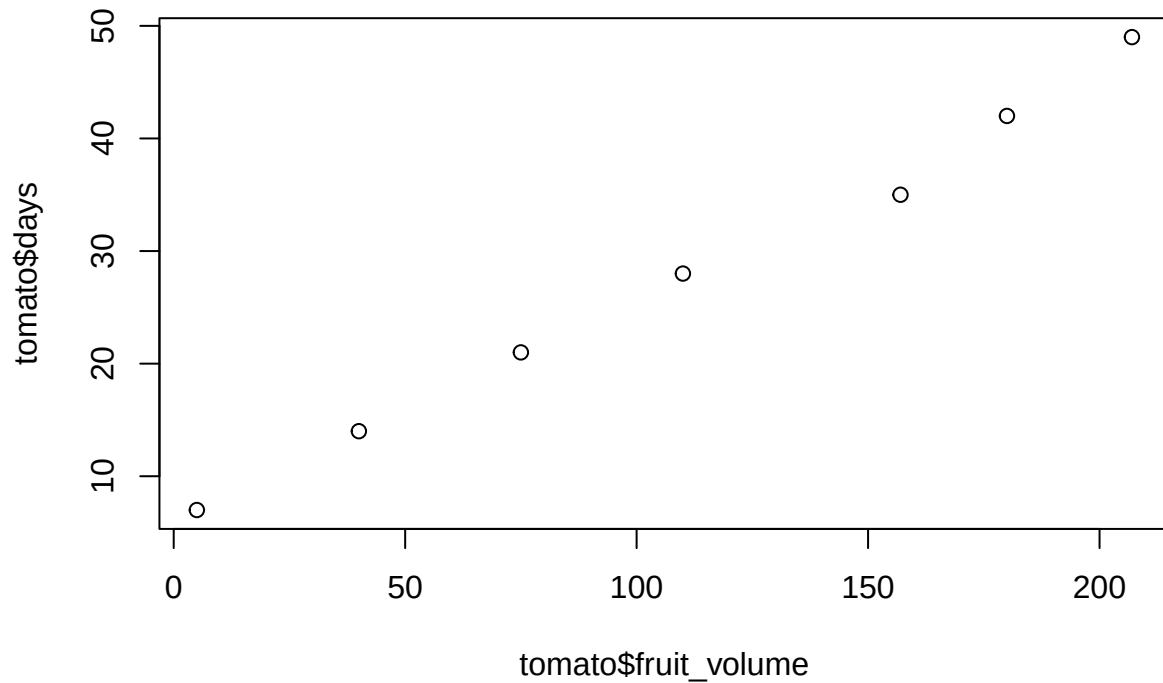
```
##   days fruit_volume
## 1    7           5
## 2   14          40
## 3   21          75
## 4   28         110
## 5   35         157
## 6   42         180
```

### Question 10

1 point

Plot `fruit_volume` as a function of `days`.

```
plot(tomato$fruit_volume, tomato$days)
```



In one sentence, describe the shape you see.

## Question 11

**2 points**

Fit and summarize a logistic nonlinear model:

```
fruit_volume ~ SSlogis(days, Asym, xmid, scal)
```

```
model2 <- nls(fruit_volume ~ SSlogis(days, Asym, xmid, scal),  
              data = tomato)  
summary(model2)
```

```
##  
## Formula: fruit_volume ~ SSlogis(days, Asym, xmid, scal)  
##  
## Parameters:  
##      Estimate Std. Error t value Pr(>|t|)  
## Asym  217.714    13.355  16.302 8.29e-05 ***  
## xmid   27.370     1.567  17.463 6.31e-05 ***  
## scal    8.304     1.093   7.597 0.00161 **
```



```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.096 on 4 degrees of freedom
##
## Number of iterations to convergence: 0
## Achieved convergence tolerance: 5.049e-06
```

Interpret each parameter in biological terms:

1. Asym
2. xmid

## Question 12

1 point

Create a prediction grid for `days` from 7 to 49 by 1, then predict `fruit_volume` on that grid.

## Question 13

1 point

Add the fitted logistic curve to your original tomato scatterplot.

## Question 14

1 point

Fit a polynomial model to the same tomato data:

```
lm(fruit_volume ~ days + I(days^2), data = tomato)
```

```
model3 <- lm(fruit_volume ~ days + I(days^2), data = tomato)
summary(model3)
```

```
##
## Call:
## lm(formula = fruit_volume ~ days + I(days^2), data = tomato)
##
## Residuals:
##      1      2      3      4      5      6      7
##  2.6667 -1.4286 -3.7143 -4.1905  9.1429  0.2857 -2.7619
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -38.57143    8.94579  -4.312  0.01253 *
## days         5.97279    0.73239   8.155  0.00123 **
## I(days^2)    -0.01846    0.01278  -1.445  0.22209
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 5.74 on 4 degrees of freedom  
## Multiple R-squared:  0.9961, Adjusted R-squared:  0.9941  
## F-statistic: 508.8 on 2 and 4 DF,  p-value: 1.533e-05
```

Then, in 3-4 sentences, compare polynomial vs logistic for this dataset using:

1. biological interpretability
2. plausibility of curve shape
3. model communication to an agronomy audience